VERSION
14.0



Glossary

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AMS GRA ARCHITECTURE VOLUMES

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1.0 Glossary

Term	Description
Abstract Service Bus (ASB)	Per the OMS Standard, the ASB is a composite element that supports all types of OMS Data Exchanges. The ASB includes the physical layer that is the foundation of the Abstract Service Bus (ASB)'s other sub-elements. It is important to note that the ASB is a logical construct, not a single physical element distinct from other Open Mission Systems (OMS) elements, and is defined and provided as a part of an OMS Platform. The ASB (and therefore the Platform) includes components hosted on Subsystems, the OCE, and Other Mission Processing elements.
Access	In the context of Cybersecurity, access consists of the ability and means to do the following: • Communicate with or otherwise interact with a system • Use system resources to handle information • Gain knowledge of the information the system contains • Control system components and functions
Access Control	In the context of Cybersecurity, access control means the process of granting or denying specific requests for the following reasons: • Obtaining and using information and related information processing services • Entering specific physical facilities (e.g., federal buildings, military establishments, border crossing entrances, etc.)
Access Control List	In the context of Cybersecurity, an access control list is a list of permissions associated with an object. The list specifies access allowed to the object by who or what and allowed operations to perform on the object.
Adapter	In the context of Agile Mission Suite (AMS) Government Reference Architecture (GRA), a structural pattern that allows incompatible interfaces between software and hardware elements to work together by converting the supplier interface to an AMS GRA defined interface. Acts as an isolated translator between a non AMS GRA defined interface and an AMS GRA defined interface allowing objects with incompatible interfaces to collaborate.
Agile Mission Suite (AMS) Government Reference Architecture (GRA) Adapter	A specialized adapter used to interface with an Agile Mission Suite (AMS) Government Reference Architecture (GRA) Service. An AMS GRA Adapter can support one or many AMS GRA defined interfaces except OMS interfaces (see Open Mission Systems (OMS) Adapter).

Term	Description
Agile Mission Suite (AMS) Government Reference Architecture (GRA) Operating System (OS) Façade Application Programming Interface (API)	The AMS GRA OS Façade API is a subset of the OMS OS Facade API, where exclusions are defined by the AMS GRA and provided in the Software Architecture Volume.
Agile Mission Suite (AMS) Government Reference Architecture (GRA) Service	An AMS GRA Service is a Level 1 MPU that implements MFP layer logic, and that interacts with other Minimum Procurable Units (MPUs) over the ABB (e.g., MFAs). AMS GRA Services may utilize GPP, GPU or FPGA Processing Components on the DPPs to which they are deployed. Capabilities enabled through use of AMS GRA Services include: • Access to MFA MPUs over the MEL interface • Communication with other AMS GRA Services over interfaces supported by the Avionics Backplane Bus (ABB) • Parallel processing through decomposition into DPs that run on multiple GPP, GPU, or FPGA Processing Components on one or more DPPs • Development by third parties through use of DPPs DKs
Aircraft Systems	Aircraft Systems refers generally to vehicle control, hydraulic, power generation and cooling, and related infrastructural elements of a flyable vehicle which are not included in the Mission System.
Anti-Tamper (AT)	Per the Anti-Tamper Engineering Framework (ATEF), the purpose of Anti-Tamper (AT) is to Deter, Impede, Detect, and Respond to attempts at hands-on reverse engineering and exploitation of systems that contain Critical Program Information (CPI).
Aperture	An aperture is an interface that converts between free space radiation and data within the system.
Architecture Element	Per the OMS Standard, OMS defines the Architecture Elements as the five major building blocks of the OMS architecture: Mission Packages, Platforms, Subsystems, Services, and Isolators.
Array	An array is a collection of sensor components, spatially distributed within an MFA face. Examples include linear or rectangular grids of Radio Frequency (RF) elements within a face on an RF MFA, or optical elements of a focal plane within an IR MFA face.
Assurance	In the context of Cybersecurity, assurance is a measure of confidence that the security features, practices, procedures, and architecture of an information system accurately mediates and enforces the security policy.

Term	Description
Attack Surface	In the context of Cybersecurity, attack surface is the collection of possible entry points for an attacker. These represent the means by which an attacker may introduce malware into the OMS instantiation or access vulnerabilities or malware that already introduced within the OMS instantiation.
Authentication	Authentication is the process of using a cryptographic method to verify the integrity and source of data.
Authoritative Reference	An authoritative reference is a source that is recognized by the consortium to be the valid content because it is considered to be accurate.
Authorization	In the context of Cybersecurity, authorization is the access privileges granted to a user, program, or process or the act of granting those privileges to perform functions and access data.
Availability	In the context of Cybersecurity, availability is the property of being accessible and useable upon demand by an authorized entity.
Avionics Backplane Bus (ABB)	The ABB supports communication between the MFA layer and the MFP layer as well as within the MFP layer.
BLACK	Designation applied to information systems, and to associated areas, cir ts, components, and equipment, in which national security information is encrypted or is not processed.
Board Support Package (BSP)	A BSP contains all low-level software items that are responsible for exposing board-level interfaces.
Boundary	The partition that logically separates internal and external communication.
Build Infrastructure	Build infrastructure provides the environment and/or tools that enable development of an MPU. The Development Kit (DK) shall provide a build environment with build infrastructure (e.g.,GNU Compiler Collection (GCC), Clang, Make, Git, Doxygen, Ansible, Python) pre-installed if required for development.
Capability	Per the OMS Standard, a Capability is a performable action useful in accomplishing missions of an OMS system. Capabilities are typically tasked, discovered, commanded, or any combination thereof, using OMS Messaging. Capabilities are documented in a non-proprietary Service Contract.

Term	Description
Classified Data Erase	Per the OMS Standard, in the context of the OMS CommandType field of the `SubsystemEraseCommand`, commanding of CLASSIFIED DATA ERASE removes all classified data created during Subsystem operation, and returns “the Subsystem to its default/native/base classification level”. Commanding CLASSIFIED DATA ERASE may occur as a part of the Subsystem shutdown command or may be invoked during normal Subsystem operation. A Subsystem will document its support for CLASSIFIED DATA ERASE in its Subsystem Design Document and/or Service Contract, and should reject a CLASSIFIED DATA ERASE command for situations it does not support. Restoring power to a Subsystem shutdown with CLASSIFIED DATA ERASE restarts the Subsystem to normal operation. See also potentially related terms: ZEROIZE and DISABLE.
Commercial Standards	Commercial standards are publicly available technical standards created by the commercial industry and neither owned nor controlled by the USG.
Commodity Tool	A commodity tool is a readily-available tool that is used as an input in the production of an AMS GRA instantiation (e.g., COTS, GOTS, OSS, etc.).
Common Interface	A Common Interface has the following attributes: • Is non-proprietary; • Exists between MPUs, or between MPUs and the platform (i.e., is external to the MPU); • Is defined by AMS GRA; • Includes documentation sufficient to enable implementation; and • Provides documentation that is available to any user of the AMS GRA without restriction.
Communications Security (COMSEC)	COMSEC is a component of Information Assurance (IA) that deals with measures and controls taken to deny unauthorized persons information derived from telecommunications and to ensure the authenticity of such telecommunications. COMSEC includes crypto security, transmission security, emissions security, and physical security of COMSEC material.
Component	A component is a generic term for a sub-assembly that forms part of a Level 2 MPU (e.g., FPGA chips, power amplifiers, network physical chips, etc.).

Term	Description
Compromise	In the context of Cybersecurity, compromise is the disclosure of information to unauthorized persons, or a violation of the security policy of a system in which unauthorized intentional or unintentional disclosure, modification, destruction, or loss of an object may have occurred.
Compute Cluster Farm (CCF)	A CCF is a physical instantiation of AMS GRA processing and Level 2 and Level 3 MPUs.
Confidentiality	In the context of Cybersecurity, confidentiality is the property of not disclosing information to unauthorized system entities (users, processes, devices) unless authorized to access the information.
Container Framework	A container framework transparently isolates processing resources managed by an operating system to enable segregated runtime environments that share operating system APIs (i.e., execute on the same kernel).
Coordinated Universal Time (UTC)	UTC - Primary time standard by which the world regulates clocks and time - it is synonymous with the (now unofficial) GMT designation. It differs from TAI in that it includes "leap seconds" to account for slowing of the Earth's rotation on its axis.
COSITE	COSITE refers to the ownship instance of antenna-to-antenna RF interference.
Countermeasures Manager (CMM)	The CMM is a Level 2 MPU that facilitates the translation and transfer of CMs technique data from AMS GRA elements to platform CMs elements.
Critical Abstraction Layer	Per the OMS Standard, a Critical Abstraction Layer is an OMS architecture concept and approach supporting the standardized OMS Messaging between the OMS Platform and the Subsystems, Services, and Isolators that are part of an OMS Mission Package. A Critical Abstraction Layer is realized by a Critical Abstraction Layer Implementation that supports the Critical Abstraction Layer API used by Subsystems, Services, and Isolators. The Critical Abstraction Layer isolates Subsystem, Service, and Isolator application software from Platform specific Critical Abstraction Layer Implementations. A Platform will include one or more Critical Abstraction Layer Implementations that allow Subsystem, Services, and Isolators to communicate using OMS Messages. The Critical Abstraction Layer interface aids portability of Subsystem, Services, and Isolators among OMS Mission Packages. The Critical Abstraction Layer allows applications to interact (get, set, send, receive, etc.) with message objects regardless of the underlying object implementation.

Term	Description
Cross Domain Solution (CDS)	A CDS is an MPU that enforces the information flow policy required to support transfer of unencrypted data between security enclaves.
Cryptographic Unit	The Cryptographic Unit is an MPU that provides cryptographic (crypto) functionality. Cryptographic security functions may include: cryptographic key management, transformation (i.e., encryption or decryption) of both data streams and files at rest, digital signature generation and verification, and secure hash generation.
Custom Tool	A custom tool is a tool that is specially purpose built for the production of an AMS GRA instantiation or DK.
Cyber Resiliency	Per the OMS Standard, Cyber Resiliency is the ability to anticipate, withstand, recover from, and adapt to adverse conditions, stresses, attacks, or compromises on systems that use or are enabled by cyber resources (National Institute of Standards and Technology (NIST) SP 800-160 vol 2).
Cybersecurity	Per the OMS Standard, Cybersecurity refers to the prevention of damage to, protection of, and restoration of computers, electronic communications systems, electronic communications services, wire communication, and electronic communication, (including information contained therein) to ensure its availability, integrity, authentication, confidentiality, and nonrepudiation.
Data at Rest (DaR)	DaR refers to data residing on persistent media where data remains on the media with power removed.
Data Plane	The data plane refers to the logical network used for transporting data.
Defense-in-Depth	In the context of Cybersecurity, Defense-in-Depth is an Information Security strategy integrating people, technology, and operations capabilities to establish variable barriers across multiple layers and missions of the organization.
Deployment Resource Management	Deployment Resource Management is an aspect of Resource Management that addresses the processing and network (e.g., ABB) resources used to host services and transfer high-bandwidth, low-latency data between the MFP services and MFAs, and among MFP services.
Detection Data	Detection data refers to the moving target indicator data produced by a sensor. Detection data can be streamed or file-based.
Development Kit (DK)	A DK contains the tools and artifacts necessary to develop, unit test, and package software for eventual deployment to GPPs, GPUs, and FPGAs. DPP and MFA MPUs include DKs as part of procurement.

Term	Description
Development Operations (DevOps)	Development Operations refers to a collection of tools and processes implementing a CI/CD development/security/operations pipeline that flows seamlessly into the operations environment by way of automated tools and processes.
Development, Security, and Operations (DevSecOps)	DevSecOps refers to a collection of tools and processes implementing a CI/CD Development Operations pipeline that also integrates security concerns into the development lifecycle.
Digital Payload (DP)	A DP is the smallest unit of deployable software on a single DPP. A Service is composed of one or more DPs, and at least one DP must expose the Service's open interface. When multiple DPs compose a service, they may be co-located on the same DPP or distributed across multiple DPPs. The DPs may communicate to each other using internal, non-Exposed interfaces.
Digital Payload Processor (DPP)	A DPP is a Level 2 MPU, typically in the form of a board, card, or module. DPPs are hosted inside a PCaB. A DPP contains one or more Processing Components for hosting service software.
Digital Payload Processor Compute Environment	The DPP Compute Environment is the collection of Processing Components and infrastructure software that together provide the environment for hosting DPs on a DPP.
Digital Signature	A digital signature is a cryptographic process used to assure data object originator authenticity, data integrity, and time stamping for prevention of replay.
Disable	Per the OMS Standard, in the context of the OMS CommandType field of the `SubsystemEraseCommand` and `SubsystemStateCommand` messages, DISABLE invokes Subsystem-specific action requiring OEM or depot-level maintenance to return the Subsystem to normal operation. DISABLE optionally may be interpreted to include the actions of ZEROIZE and CLASSIFIED DATA ERASE. If it does not, it could be required to invoke CLASSIFIED DATA ERASE and/or ZEROIZE in addition to DISABLE in order to address all classified data. A Subsystem will document its support for DISABLE in its Subsystem Design Document and/or Service Contract. See also potentially related terms: CLASSIFIED DATA ERASE and ZEROIZE.
Distributed Guaranteed Time (DGT)	DGT is a time synchronization and distribution protocol supported by AMS GRA.

Term	Description
Domain	In the context of Cybersecurity, domain is an environment or context that includes a set of system resources and a set of system entities that have the right to access the resources as defined by a common security policy, security model, or security architecture (see also Security Domain).
DPP API	The Digital Payload Processor (DPP) Application Programming Interface (API) aggregates the interfaces available to AMS GRA and OMS Services on a DPP to interact with General Purpose Processor (GPP), Field-Programmable Gate Array (FPGA), and Graphics Processing Unit (GPU) Processing Components. The GPP DPP API is the AMS GRA Operating System (OS) Façade API. This concept leverages the OMS OS Façade API, but allows for exclusions defined by the AMS GRA. Using the OMS-defined list of disallowed Portable Operating System Interface (POSIX) calls as a base, the AMS GRA reserves the right to exclude calls from that list and allow them, according to architecture-defined standards. These exclusions are provided below. The AMS GRA has not yet defined any exclusions to the OMS OS Façade API. Of the three DPP types (FPGA, GPU, GPP), only the FPGA API is entirely defined by the DPP DK. The GPU and GPP DPP APIs have some API definitions required by the DPP DK, but can have additional API requirement leverages to one or both of them by the Mission System Integrator (MSI). An example of an additional API that an MSI could leverage would be mandating OpenGL support.
Dwell	Dwell refers to a period of time allocated in an antenna schedule for specific processing. There can be receive dwell, transmit dwell, or both Rx / Tx dwells. Multiple independent and concurrent dwells can be executed limited by the capability of the receiver or MFA processor.
Dynamic Authorization	A technology in which authorization and access rights to a network, applications, data, or other sensitive assets are granted and modifiable at run-time using attribute-based policies.

Term	Description
Element	Element is a generic term for any construct in AMS GRA that is instantiable or to any instantiation of that construct. The intent is to allow a term to refer to instantiable things in the architecture in a context that is not specific to a particular construct, e.g., Skill, MFA, Service, and DPP. This also allows the document set to avoid confusion by the use of other terms for the same purpose (for example, the term “component”). As an exception to this stylistic rule, the term “element” is permissible when referring to antenna “elements” within an aperture, since this is a long-standing term of art for those hardware components.
Enclave	In the context of Cybersecurity, an enclave is a collection of information systems connected by one or more internal networks under the control of a single authority and security policy. The structuring of the systems may be by physical proximity or by function, independent of location.
Encrypted Bus	An Encrypted Bus is a concept for providing domain separation for simultaneous processing, transmission, and storage of information running at different classifications levels. Encryption techniques are used to facilitate access to information or services in different security domains and provide confidentiality and integrity for data in transit.
Endpoint	Either the client or server side of a network connection.
Exposed Interface	An Exposed Interface has the following attributes: • Exists between MPUs, or between MPUs and the platform (i.e., is external to the MPU); • Is defined by MPU Providers or MSIs; • Includes documentation sufficient to enable implementation; and • Provides documentation that is available to specific users of the AMS GRA with restriction.
Façade	Per the OMS Standard, the Façade is an OS API that abstracts Services (applications) from the underlying OS/hardware implementation. The Façade API defines the precise set of OS calls allowed to be used by an OCE Hosted Service. A Service’s use of only the defined API accomplishes the abstraction from the underlying OS and hardware.
Firmware	Firmware is considered software if it can be deployed or changed using a DK. Firmware is considered hardware if it cannot be deployed or changed using a DK. An example of firmware is FPGA code.

Term	Description
Flexibility	Per the OMS Standard, flexibility refers to the architecture's ability to accommodate changing mission requirements. This includes the ability to accommodate different functional decompositions of behaviors, different types of data products, and varying command and control methodology to support varying aircraft and Mission System architectures.
Function	A function is an intrinsic behavior of a system that supports a capability.
General Resource Management	General Resource Management works in conjunction with Deployment and SRM to coordinate and manage resource allocation and execution for system aspects.
Guest Operating System	A guest refers to an operating system installed on a VM.
Hashing	Hashing is the process of using a mathematical algorithm against data to produce a numeric value that is representative of that data.
Hypervisor	A hypervisor is software that emulates and isolates processing resources to enable segregated diverse runtime environments to execute on the same physical processing resources. Type 1 hypervisors execute directly on the processing resources they manage, similar to an operating system. Type 2 hypervisors execute on top of an operating system.
In-phase/Quadrature (I/Q)	I/Q data are integer-valued functions used to represent RF signal energy digitally at base band.
Information Flow Policy	Information Flow Policy is a synonym for CDS policy.
Integrity	In the context of Cybersecurity, integrity is the property indicating no occurrence of unauthorized modification of an entity.
Integrity Verification	Integrity Verification refers to the confirmation, through the provision of objective evidence, that no unauthorized modification of an entity has occurred.
Interface Characteristics	Interface Characteristics refer to the complete description of the interface's normative and abnormal behavior allowing a user of the interface to implement all aspects of the mating interface successfully. Types of characteristic information may include (but are not limited to) Physical Characteristics, Electrical Characteristics, Logical Characteristics, and Environmental Characteristics.

Term	Description
Isolation	Per the OMS Standard, isolation occurs in the OMS architecture in two ways. First is the use of an Isolator that limits the effects of change between the Mission Package boundary and another system. The second is providing physical isolation to ensure complete separation from safety-critical and nuclear surety.
Key	A key is a numerical value used to control cryptographic operations, such as decryption, encryption, signature generation, or signature verification.
Key Management	Key management refers to the activities involving the handling of cryptographic keys and other related security parameters (e.g., Initialization Vectors and passwords) during the entire lifecycle of the keys, including their generation, storage, establishment, entry and output, and zeroization.
Layers	Layers share data processing and timing tiers that control coupling, ensure timeliness of response, and partition processing needs (e.g., MFA layer, MFP layer, MPS layer, and Platform layer).
Least Privilege	Least Privilege refers to the principle that the design of a security architecture should include granting each entity the minimum system resources and authorizations necessary to perform its function.
Line Replaceable Unit (LRU)	A LRU is an essential support item removed and replaced at field level to restore an end item to an operationally ready condition.
Link-16	Per the OMS Standard, Link-16 is a military tactical Data Exchange Network created and used by the United States and adopted by some of its Allies and by North Atlantic Treaty Organization.
Local Function	A Local Function is a Level 1 MPU existing only in the MFA layer, exposed to a developer through a DK. Local Functions are not AMS GRA or OMS Services.
Logical Isolation	Logical isolation refers to the separating and isolating of data at different classification levels using software/firmware controls on same hardware (i.e., MLS).
Minimum Procurable Unit (MPU)	MPUs are Mission Systems elements within AMS GRA that can be independently competed after the point of manufacture with the intent of rapid capability and agile upgrade.
Mission	Per the OMS Standard, a mission refers to a period of time that includes pre-mission planning and mission execution, but excludes post-mission activities.

Term	Description
Mission Agnostic Service Infrastructure (MASI)	The MASI is the set of services responsible for tasks such as startup, health assessments, deployment, and logging. These services support the operation of Skills.
Mission Agnostic Skills	Mission Agnostic Skills are Skills that provide Common functionality (i.e., support multiple other skills in a standard way) and mission invariant functionality (i.e., do not change function across mission types).
Mission Data File	Per the OMS Standard, a Mission Data File is a configuration file or reference data used by a Subsystem or Service which is required for the execution of the mission.
Mission Package	Typically when capitalized, Mission Package refers to OMS Mission Package.
Mission Platform	A Mission Platform is a mission-capable vehicle system which may act as a part of a Weapon System and should be interpreted as aligned with an Air Vehicle at level 2 in traditional DoD requirements derivation. When standing alone, it refers to an AMS GRA-adopting Mission Platform instantiation.
Mission Processing Subsystem (MPS)	The MPS is a logical construct and not an MPU by itself. An MPS is a collection of DPPs, MLS/NVM Data Storage, and Security Modules that provide a General Processing capability. The MPS is allowed to exchange information over the ASB only. The hardware building blocks of an MPS are the same as an MFP; the difference lies in the types of capabilities implemented.
Mission Specific Skills	Mission Specific Skills are a subset of Skills that add mission specific capability to the system. This set of Skills is likely to vary between platform types due to the types of supported missions.
Mission System	A Mission System is a combination of Level 1, Level 2, and Level 3 MPUs providing the mission capabilities for a Mission Platform. When standing alone, it refers to an AMS GRA-compliant instantiation of a Mission System.
Mission System Integrator (MSI)	The MSI is the party responsible for the overall integration of a compliant Mission System instantiation. An MSI operates under the guidance of the Procuring Authority.
Multi-Function Aperture (MFA)	An MFA is a logical composition which may consist of analog and digital electronics, firmware and software (i.e., OEM Functions), and processing resources (i.e., MFA Compute). An MFA serves as the source of interaction between the free space environment and the Mission System. All MFAs have one or more MFA Face(s); however, an MFA may have physical components distributed across the Mission Platform.

Term	Description
Multi-Function Aperture (MFA) Compute	MFA Compute is the set of processing resources (i.e., GPPs, GPUs, and FPGAs) associated with an MFA.
Multi-Function Aperture (MFA) Compute Interface	The MFA Compute Interface allows OEM Functions and Local Functions to interact with the processing resources of the MFA Compute. When the MFA Compute is instantiated within the physical MFA, the Compute Interface is an Open interface defined by the MFA provider. When the MFA Compute is instantiated within a DPP dedicated to the MFA, the Compute Interface is the DPP API.
Multi-Function Aperture (MFA) Encapsulation Layer (MEL)	The MEL is included in AMS GRA to define the Common and Open set of interfaces and patterns of interaction associated with any AMS GRA compliant Multi-Function Aperture (MFA). The MEL specifically defines the non-proprietary, Common interface between MFP hosted Services and the MFA.
Multi-Function Aperture (MFA) Face	An MFA Face is a sub-assembly of an MFA, located at or near the outer mold line, which has a single, contiguous Field of Regard (FoR). For example, a single Type 3 MFA may provide Forward Left, Forward Right, and Aft Faces as sub-assemblies.
Multi-Function Aperture (MFA) Internal Data	MFA Internal Data is the set of data interfaces provided by the MFA OEM for Local Functions to utilize. These data interfaces are defined by the Local Function DK delivered with the MFA.
Multi-Function Aperture (MFA) Local Function Development Kit	An MFA Local Function DK that allows a developer to create a Local Function.
Multi-Function Aperture (MFA) Type 1	MFA Type 1 is an MPU that provides self-contained programmable transmit and/or receive capability. The MFA is self-contained in that it does not use external processing hosted on an MFP to implement the capabilities provided.
Multi-Function Aperture (MFA) Type 2	MFA Type 2 is an MPU that provides programmable transmit and/or receive capability. Type 2 MFAs, at a minimum, provide digitized I/Q data to external processing. They may also provide processed data in various forms, including PDW data. Type 2 MFAs transmit signals as directed by externally-hosted capabilities and/or local modes.

Term	Description
Multi-Function Aperture (MFA) Type 3	MFA Type 3 is an MPU that provides programmable transmit and/or receive capability through one or more faces. Type 3 MFAs must provide internal scheduling capable of interleaving multiple scheduler requests among controlled apertures. Type 3 MFAs, at a minimum, provide digitized I/Q data to external processing. They may also provide processed data in various forms, including PDW data. Type 3 MFAs transmit signals as directed by externally-hosted capabilities and/or local modes.
Multi-Function Aperture (MFA) Type 4	MFA Type 4 is an MPU that provides programmable transmit and/or receive capability through one or more faces. Type 4 MFAs must provide internal scheduling capable of interleaving multiple scheduler requests among controlled apertures. Type 4 MFAs, at a minimum, provide digitized frame or pixel data to external processing. They may also provide processed data in various forms including detection data. Type 4 MFAs transmit signals as directed by externally-hosted capabilities and/or local modes.
Multi-Function Aperture (MFA) Type 5	MFA Type 5 is an MPU that provides programmable transmit and/or receive capability through one or more faces. Type 5 MFAs, at a minimum, provide digitized frame or pixel data to external processing. They may also provide processed data in various forms, including detection data. Type 5 MFAs transmit signals as directed by externally-hosted capabilities and/or local modes.
Multi-Function Processor (MFP)	The MFP is a logical construct and not an MPU by itself. An MFP is collection of DPP, Data Storage, and PCaB MPUs that provide a Signal Processing capability. The MFP is allowed to exchange information over the ASB and ABB.
Multi-Level Security (MLS)	MLS refers to the concept of providing simultaneous processing, transmission, and storage of information at different classifications levels using data tagging techniques to facilitate access to information or services in different security domains.
Multilevel Mode	In the context of Cybersecurity, multilevel mode is a mode of operation wherein all of the following statements are satisfied concerning the users who have direct or indirect access to the system, its peripherals, remote terminals, or remote hosts: • Some users do not have a valid security clearance for all the information processed in the information system. • All users have the proper security clearance and appropriate formal access approval for that information to which they have access. • All users have a valid need-to-know only for information to which they have access.

Term	Description
Non-Prescriptive Content	Non-prescriptive content includes but is not limited to elements that provide supplemental information, design guidance, descriptions of rationale of architectural decisions, and reference implementations.
Normative	Normative refers to the standard method of behavior (e.g., in performing an operation, providing a status, etc.). Non-normative is an alternate method available but not used by default.
Object	In the context of Cybersecurity, an object is a passive system-related entity (e.g., devices, files, records, tables, processes, programs, domains) containing or receiving information. Access to an object implies access to the information it contains.
Open Addition	An Open Addition adds capability to areas of the Common MEL API that are not intended to be extended/specialized for the purpose of providing additional MFA functionality to a Service on the MFP-layer. An Open Addition can take the form of a derived class, new functions, or new base classes. Open Additions are non-proprietary and include documentation sufficient to enable implementation. Adding new parameters to existing functions is not allowed because of the risk of collisions, even though it is technically possible. The following Common MEL API classes are intended to be extended/specialized, and derived classes from these classes are not considered Open Additions: Weights, ModulationExtensionBase.
Open Compute Environment (OCE)	The OCE is the portion of the DPP Compute environment that provides Services with an interface to operating system facilities on a GPP Processing Component. A POSIX implementation is an example of an OCE. Per the OMS Standard, the OCE defines a computing environment on which OMS Services, Isolators, or Adapters can run. A Platform is required to provide an OCE, but Services, Isolators, and/or Adaptors are not required to use a Platform's OCE. If a Service/Isolator/Adapter requires software hosting by a Platform but does not use the Platform OCE, then all necessary Platform support must be documented in the applicable Service/Isolator/Adapter OMS-required documentation. The intent of the OCE is to provide an environment that will facilitate Service, Isolator, and Adapter portability between OMS Platforms and provide a common target for the host environment of OMS Services.

Term	Description
Open Interface	An Open Interface has the following attributes: • Is non-proprietary; • Exists between MPUs, or between MPUs and the platform (i.e., is external to the MPU); • Is defined by MPU Providers or MSIs; • Includes documentation sufficient to enable implementation; and • Provides documentation that is available to any user of the AMS GRA without restriction.
Open Mission Systems (OMS)	OMS is a government-owned architecture specification that focuses on the interfaces between software services and hardware subsystems, and how that data is exchanged across those interfaces. The AMS GRA utilizes OMS for interfaces at the MPS level.
Open Mission Systems (OMS) Adapter	Per the OMS Standard, an OMS Adapter refers to the hardware and software elements required to enable a particular non-OMS subsystem or non-OMS service to support OMS Data Exchanges with a particular instantiation of an OMS Platform. In the case of a non-OMS subsystem/service being connected into an OMS Platform, the Adapter provides command and control translations required to enable the subsystem/service to support OMS Data Exchanges. An Adapter also provides any required physical transport, protocol and message translations (which would include using the Critical Abstraction Layer for Subsystems that do not natively use the Critical Abstraction Layer).
Open Mission Systems (OMS) Data Exchanges	Per the OMS Standard, OMS Data Exchanges are Data Transfers, OMS Messages, Security Information Exchanges, and Special Signals.
Open Mission Systems (OMS) Data Transfer	Per the OMS Standard, OMS Data Transfer is a Data Transfer which is one of the four top-level information exchange types described by OMS. OMS Data Transfer deals with file and product exchanges that are restricted to the defined data types and their allowed data formats, transfer protocols/APIs, and sharing patterns described by the OMS Standard. An OMS Data Transfer must adhere to the following restrictions: 1) does not overlap with OMS Messages, 2) does not overlap with any other OMS Data Exchanges, and 3) does not transfer external standards second hand.

Term	Description
Open Mission Systems (OMS) Isolator	Per the OMS Standard, an OMS Isolator is an OMS Architecture Element that is only comprised of one or more software components. An Isolator operates between an OMS Mission Package and elements external to the OMS Mission Package; its OMS-facing interface and predefined constraints and policies are specified by a non-proprietary Service Contract. An Isolator separates OMS Data Exchanges on the OMSPlatform from elements external to the OMS Mission Package with the express purpose of preventing unwanted interactions and cross-coupling interference, while at the same time providing the means for the OMS Architecture Elements of a Mission Package to interact with external elements.
Open Mission Systems (OMS) Messages	Per the OMS Standard, an OMS Message is a unit of information sent or received using an OMS Critical Abstraction Layer. The content and structure of OMS Messages are defined using an eXtensible Markup Language (XML) Schema Definition (XSD) specification that conforms to the UCI Schema Style and Design Specification (OAC-SPC-001). OMS Messages are primarily command-and-control-oriented exchanges used inside of a Mission Package to coordinate key mission functionality, technical capabilities, and operational behaviors of the interaction between the Architecture Elements (Platform, Subsystems, Services, and Isolators).
Open Mission Systems (OMS) Mission Package	Per the OMS Standard, an OMS Mission Package is an OMS Architecture Element and is the highest level OMS construct. A Mission Package consists of a Platform, Subsystems, Services, and Isolators, as documented in the Mission Package Worksheet.
Open Mission Systems (OMS) Platform	Per the OMS Standard, an OMS Platform is an OMS Architecture Element, which is itself a composite architecture. A Platform includes the ASB, mission processing (including an OCE), and typically provides Time and Position information for the Mission Package. The Platform is assessed in accordance with the non-proprietary Platform Description Document (PDD). A Platform will include non-OMS software that provides a measure of robustness for the underlying infrastructure. One example is infrastructure software and settings necessary to protect information passing through the ASB. A second example is software that manages the underlying infrastructure such as the network that forms the ASB, the processing elements that form mission processing (including the OCE), and data storage.

Term	Description
Open Mission Systems (OMS) Security Information Exchange	Per the OMS Standard, Security Information Exchange are one of the four top-level information exchange types described by OMS (the other three types are: OMS Messaging, Data Transfer, and Special Signals). Security Information Exchanges are defined in the OMS Standard and are used by special security processing and are not typically “seen” or used by mission Services (in the OCE) or Services in a Subsystem.
Open Mission Systems (OMS) Service	Per the OMS Standard, an OMS Service is an OMS Architecture Element that is only comprised of one of more software components. A Service provides mission functions and/or Capabilities that are accessed in accordance with a prescribed interface and predefined constraints and policies specified by a non-proprietary Service Contract.
Open Mission Systems (OMS) Special Signal	Per the OMS Standard, Special Signals are one of the four top-level information exchange types described by OMS (the other three types are: OMS Messaging, Data Transfer, and Security Information Exchanges). These signals are used where performance (e.g., latency) or safety/security concerns preclude the use of OMS Messaging. Often Special Signals deal with demanding latency and safety requirements that are not compatible with messaging software overhead and reliability. The OMS Standard describes and provides guidance for Special Signals.
Open Mission Systems (OMS) Subsystem	Per the OMS Standard, an OMS Subsystem is an OMS Architecture Element that is comprised of one or more hardware and software components. A Subsystem provides mission functions and/or Capabilities. A Subsystem is accessed in accordance with a prescribed interface and predefined constraints and policies specified by a non-proprietary Service Contract and a non-proprietary Subsystem Description Document.
Openness	AMS GRA uses the term Openness as a generic description for the interface categories associated with the MPUs. The defined levels of Openness (from most to least desirable) are Common, Open, and Exposed.
Original Equipment Manufacturer (OEM) Function	The OEM Function is software or firmware that is specific to an MFA and developed by the OEM. It does not support any third-party replacement.
Pass-Through Interface	An interface that allows data to pass between endpoints relatively instantly without being processed, filtered, interacted with or modified.

Term	Description
Physical Isolation	Physical isolation refers to the separating and isolating data at different classification levels using physical hardware instantiations.
Platform	Typically when capitalized, Platform refers to the AMS GRA Platform.
Prescriptive Content	Prescriptive content refers to content in the AMS GRA Product Set that is necessary to direct and assess the architectural compliance of a Mission System instantiation and its constituent MPUs. Prescriptive content may include criteria for compliance to AMS GRA, definitions and descriptions of architectural elements, interface definitions, references to industry standards, and AMS GRA glossary definitions.
Processing Component	A processing component is a sub-assembly of a DPP that can host/run service software DPs, enabled by the DPP DK. Processing components may be GPP Components, FPGA Components, or GPU Components.
Processor Chassis and Backplane (PCaB)	An assembly consisting of a PCaB MPU hosting supported hardware. Supported hardware is any combination of allowed PCaB-hosted MPUs and optional MFA Compute hardware. While the PCaB itself and the contained MPUs are governed by compliance requirements, the Populated PCaB is not subject to any additional requirements.
Procuring Authority	The Procuring Authority is the party responsible for the procurement of the Mission System instantiation.
Provision	Per the OMS Standard, a provision is any preparation made beforehand to deal with a future need without disrupting existing capabilities or implementations. Provisioning means to make accommodations for subsystems or services, recently added or in place of existing subsystems or services, where those accommodations enable the top-level business goals. A specific OMS example would be provisioning for special signals at a platform level when initial subsystems are Tier 1 or Tier 2, and do not use those special signal connections. By providing the special signal connections and propagation media on the platform (i.e., running the wires), integration of a Tier 3 system in place of a lower-tier system may occur without requiring changes to the platform.

Term	Description
Pulse Descriptor Word (PDW)	A PDW provides data regarding measurements and configuration details on an incoming detected energy created by signal processing. Examples of information are frequency, pulse width, amplitude, angle of arrival, modulation, time information, interoperability information, and MFA configuration information.
Resource Management	Resource Management is the process of scheduling, allocating, using, and managing the resources of the platform to maximize effectiveness while completing a mission. These include Sensor Resources, Deployment Resources, and General Resources.
Root of Trust (RoT)	A RoT is a highly reliable hardware, firmware, and/or software component that performs a specific, critical security function such as secure key distribution, or secure key storage.
Sanitization	Sanitization refers to the actions taken to render data written on media unrecoverable by both ordinary and extraordinary means.
Scenario	A scenario is a sequence of events, activities, or interactions among participants.
Secure Architecture	A high level design that includes features such that trust can be established.
Secure Boot	Secure Boot is the process of establishing a chain of trust from power on through capability execution on a computing element.
Secure Communication Protocol	An open communication protocol that provides the appropriate confidentiality, authentication, authorization, and content-integrity protection.
Secure Communications	Secure communications refers to the protection of data from unauthorized access, modification, or destruction during transmission.
Secure Communications Channels	A secure communication channel is a dedicated data transmission path that ensures confidentiality, integrity, and authenticity of data transmitted between two nodes in a network.
Security Attributes	In the context of Cybersecurity, security attributes are an abstraction representing the basic properties or characteristics of an entity with respect to safeguarding information; typically associated with internal data structures (e.g., records, buffers, files) within the information system which are used to enable the implementation of access control and flow control policies; reflect special dissemination, handling, or distribution instructions; or support other aspects of the information security policy.

Term	Description
Security Functions	In the context of Cybersecurity, security functions are the security-related features, functions, mechanisms, services, procedures, and architectures implemented within organizational information systems or the environments in which those systems operate. Security functions also refers to an information system's hardware, software, and/or firmware that are responsible for enforcing the system security policy and supporting the isolation of code and data on which the protection is based.
Security Module (SM)	A SM is a logical construct composed of a CDS or a Cryptographic MPU in the AMS GRA. SMs provide certifiable security-related processing, for elements of the AMS GRA. SMs are housed within a PCaB.
Security Policies	Security policies refer to the trusted configuration of the system security elements that provides the confidentiality, integrity, and availability of the security enclaves. In the context of CDS, security policies are a ruleset for a Transfer CDS that defines how/when/what a Transfer CDS must audit.
Sensor Resource Management (SRM)	SRM is an aspect of Resource Management that addresses the run time operation and optimization of Skills using MFAs, as well as managing RF spectrum across both AMS GRA apertures and other platform emissions.
Service	A Service is an MPU of software implementing one or more functions and/or behaviors and having a defined interface to the ASB and/or ABB. Services are units of software that can aggregate into one or more Skills, but each Service exists within a single layer (e.g., MFP or MPS). A Service is capable of deployment on one or more DPPs.
Service Contract	A Service Contract is a document that defines and describes the information and behaviors required by a Service, Subsystem, or Isolator to support service message exchange along with its pedigree. The Service Contract contains a complete description of the Service/Subsystem/Isolator such that an application interfacing with the Service, Subsystem, or Isolator can invoke the Service/Subsystem/Isolator and interact with it successfully via applicable message sets (i.e. OMS Messages). Unit testing results that demonstrate the Service/Subsystem/Isolator operates in accordance with the Service Contract are also referenced/included in the Service Contract.

Term	Description
Skill	A Skill is an aggregation of AMS GRA Services, OMS Services, and/or Local Functions that provide a desired effect and may span MFA, MFP and MPS layers. If a Skill consists of multiple Services, each Service must be capable of managed deployment and documented to support replacement by a third party. A Skill is a Procurable Unit.
Software Assurance	Software assurance refers to the process, tools, and techniques used to design, develop, analyze, and test software applications to establish a level of confidence that security features accurately mediate and enforce the security policy of the information system.
Software Bill of Materials	A Software Bill of Materials is an inventory of all software components (proprietary and open source), open source licenses, and dependencies in a given MPU. A Software Bill of Materials provides visibility into the software supply chain and any license compliance, security, and quality risks that may exist.
Stream	A stream consists of digitized data flow to/from an MFA through the data plane. There may be one or more streams to/from an MFA. There are different types of data streams. For example, I/Q data is a stream. Filtering of the stream may occur, but it is otherwise a direct representation of free space fields. PDW data is a stream.
Subarray	A subarray is a subset of elements of an MFA.
Subsystem	Typically when capitalized, Subsystem refers to OMS Subsystem (see definition of OMS Subsystem for more information).
Supply Chain Risk Management	Supply Chain Risk Management is a systematic process for managing supply chain risk by identifying susceptibilities, vulnerabilities, and threats throughout the DoD's "supply chain" and developing mitigation strategies to combat those threats whether presented by supplier, the supplied product and its subcomponents, or the supply chain.
Susceptibility	In the context of Cybersecurity, susceptibility is any intentional (designed in) or unintentional weakness, feature, or situation that could potentially assist an adversary in conducting a cyber attack.
System High	System High represents the highest level of security classification supported by an information system.
TEMPEST	U.S. NSA specification and a NATO certification referring to spying on information systems through leaking emanations, including unintentional radio or electrical signals, sounds, and vibrations.
Temps Atomique International (International Atomic Time) (TAI)	TAI - High precision atomically sourced time standard related to UTC but does not include leap seconds.

Term	Description
Threat	A threat is any circumstance or event with the potential to adversely impact organizational operations (including mission, functions, image, or reputation), organizational assets, individuals, other organizations, or the nation through an information system via unauthorized access, destruction, disclosure, modification of information, and/or denial of service.
Threat Source	A threat source refers to an intent and method targeted at the intentional exploitation of a vulnerability or a situation and method that may accidentally exploit a vulnerability.
Transmission Security (TRANSEC)	TRANSEC measures (security controls) are applied to transmissions in order to prevent interception, disruption of reception, communications deception, and/or derivation of intelligence by analysis of transmission characteristics such as signal parameters or message externals. Note: TRANSEC is that field of COMSEC, which deals with the security of communication transmissions, rather than that of the communicated information.
Trust	Confidence that the element is in a known and documented state.
Trust Controller	A Trust Controller is a component of a logically separated multi-level system that is responsible for controlling Trust Functions.
Trust Function	A Trust Function is a component of a logically separated multi-level system that is responsible for brokering data onto/off of a logically separated multi-level bus.
Trusted Element	An element that is known to be in an approved state with C-I-A protection objectives.
Trusted Switch	A Trusted Switch is a hardware and/or software component that reliably enforces the prescribed security policy for a network.
Trusted System	A collection of trusted and untrusted elements that are able to maintain the confidence to perform a function.
Untrusted Element	An element that has not been verified to be in an approved state.
Use Case	A use case is a system modeling technique used to describe a system's behavior under various conditions.
User	In the context of Cybersecurity, a user is an individual, or (system) process acting on behalf of an individual, authorized to access an information system.
Virtual Aperture (VA)	A VA is a configuration of a subset of MFA capability, presented as a single logical aperture to a Service. A single MEL implementation may concurrently support multiple VAs.

Term	Description
Virtual Aperture Development Tool	MFA suppliers may provide the ability for 3rd parties to define new VAs in addition to those provided by the MFA, by providing a VA Development Tool. This tool allows 3rd party developers to specify not only the public attributes of the new VA, but also the processing chains (i.e., Tx/Rx Element Groups, Endpoints, OEM Functions and Local Functions) within the MFA that implement the action of that VA. Since Local Functions must be associated with VAs via VA attributes, any MFA that allows 3rd party Local Function development must also provide a VA Development Tool.
Virtual Machine (VM)	Per the OMS Standard, a VM is a set of virtual hardware, including processor, memory, and I/O.
Virtualization	Per the OMS Standard; in general, this is the computing concept of simulating processing hardware, OS, storage devices, and/or network resources. For OMS mission processing (including an OCE), virtualization is an approach to isolate Service (application) software from changes in the underlying hardware. Virtualization technology, including virtual machines, cloud infrastructure, and containerization, enables Services to be integrated into mission processing as long as the operating system and processor architecture on which the Service was developed and tested is supported in the virtualized environment.
Vulnerability	A vulnerability is a weakness in an information system, system security procedures, internal controls, or implementation that is available for exploitation by a threat source.
World Geographic Survey (WGS-84)	WGS-84 is a reference system for geospatial location and positioning established in 1984 - Reference used by GPS.

Term	Description
Zeroize	Per the OMS Standard, in the context of the OMS CommandType field of the `SubsystemEraseCommand` and `SubsystemStateCommand` messages, ZEROIZE indicates a command to erase cryptographic keys and associated classified data in a Subsystem. Depending on the chosen interpretation of the term "and associated classified data," the ZEROIZE by itself (i.e., without a CLASSIFIED DATA ERASE) may either be interpreted to remove all classified data and return the Subsystem to its default classification level or remove only the portion of the classified data "associated" with the crypto data (and may not, in that case, return the subsystem to its base classification level). In this second interpretation, CLASSIFIED DATA ERASE should be combined with ZEROIZE. In either case, an organizational maintenance action will be required to reload keys and enable the Subsystem to return to normal operation. A Subsystem will document its support for ZEROIZE in its Subsystem Design Document and/or Service Contract. See also potentially related terms: CLASSIFIED DATA ERASE and DISABLE.

Table 1.0.1: The Glossary Table Details Common Terminology Used in the AMS GRA Volumes.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
API	Application Programming Interface
ASB	Abstract Service Bus
AT	Anti-Tamper
ATEF	Anti-Tamper Engineering Framework
BSP	Board Support Package
CCF	Compute Cluster Farm
CDS	Cross Domain Solution
CI/CD	Continuous Integration/Continuous Delivery
CM	Countermeasure
CMM	Countermeasures Manager
COMSEC	Communications Security
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
DaR	Data at Rest
DevSecOps	Development, Security, and Operations
DGT	Distributed Guaranteed Time
DK	Development Kit
DoD	Department of Defense
DP	Digital Payload
DPP	Digital Payload Processor
FoR	Field of Regard
FPGA	Field-Programmable Gate Array
GCC	GNU Compiler Collection
GOTS	Government-Off-The-Shelf
GPP	General Purpose Processor

GPS	Global Positioning System
GPU	Graphics Processing Unit
GRA	Government Reference Architecture
I/O	Input/Output
I/Q	In-phase/Quadrature
IA	Information Assurance
IR	Infrared
LRU	Line Replaceable Unit
MASI	Mission Agnostic Service Infrastructure
MEL	MFA Encapsulation Layer
MFA	Multi-Function Aperture
MFP	Multi-Function Processor
MLS	Multi-Level Security
MPS	Mission Processing Subsystem
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
NATO	North Atlantic Treaty Organization
NIST	National Institute of Standards and Technology
NSA	National Security Agency
NVM	Non-Volatile Memory
OCE	Open Compute Environment
OEM	Original Equipment Manufacturer
OMS	Open Mission Systems
OS	Operating System
OSS	Open Source Software
PCaB	Processor Chassis and Backplane
PDW	Pulse Descriptor Word
POSIX	Portable Operating System Interface
RF	Radio Frequency

RoT	Root of Trust
Rx	Receive
SK	Starter Kit
SM	Security Module
SRM	Sensor Resource Management
TAI	Temps Atomique International (International Atomic Time)
TRANSEC	Transmission Security
Tx	Transmit
UCI	Universal Command and Control Interface
USG	United States Government
UTC	Coordinated Universal Time
VA	Virtual Aperture
VM	Virtual Machine
XML	eXtensible Markup Language

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision adds prose updates and clarifications.	July 24, 2025	SK13.0
B	This revision removes instances of "Facade" and related prose in 3 places (on pages 1, 3, and 13).	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Glossary Between Documentation Releases.